

Lecture 10

Self-supervised learning (SSL)

$$\{(x_i, y_i)\}_{i=1}^L$$

↑

$$\frac{1}{L} \sum \mathcal{L}(f(x_i), y_i) \rightarrow \min$$

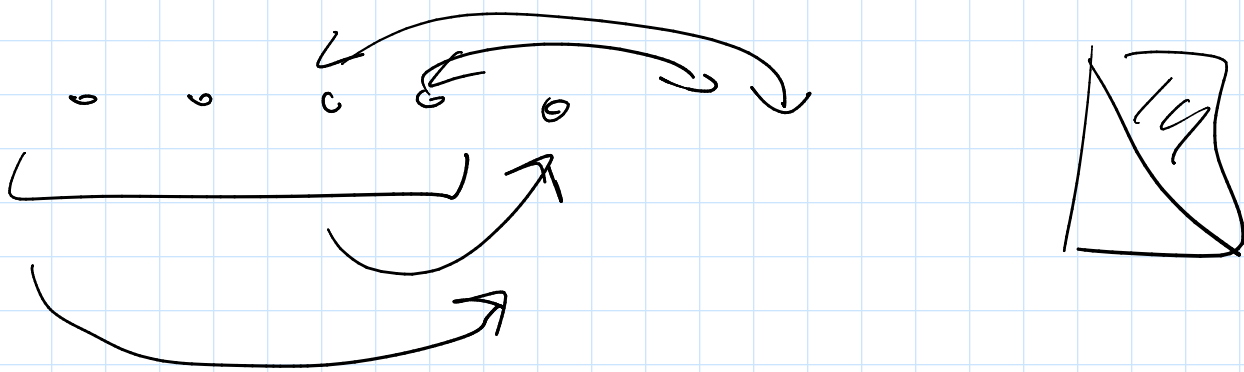
Supervised

- Word2Vec, FastText
- Auto-encoders
- Language Models

GPT (Generative Pre-training of Transformers)

Architecture — transformer decoder

Task — language modeling



BERT (Bidirectional Encoder Representations from Transformers)

Architecture — transformer encoder

Task — masked language modeling

This | is | a | sample | sentence | .

↓
[Mask] | is | a | sample | [MASK] | .

↓
Transformer
encoder

↓
Tachograph | is | a | sample | word | .
distribution over tokens

- loss only over masked tokens

All tokens

↓
15% masked

85% unchanged

80%

[MASK]

10%

random
token

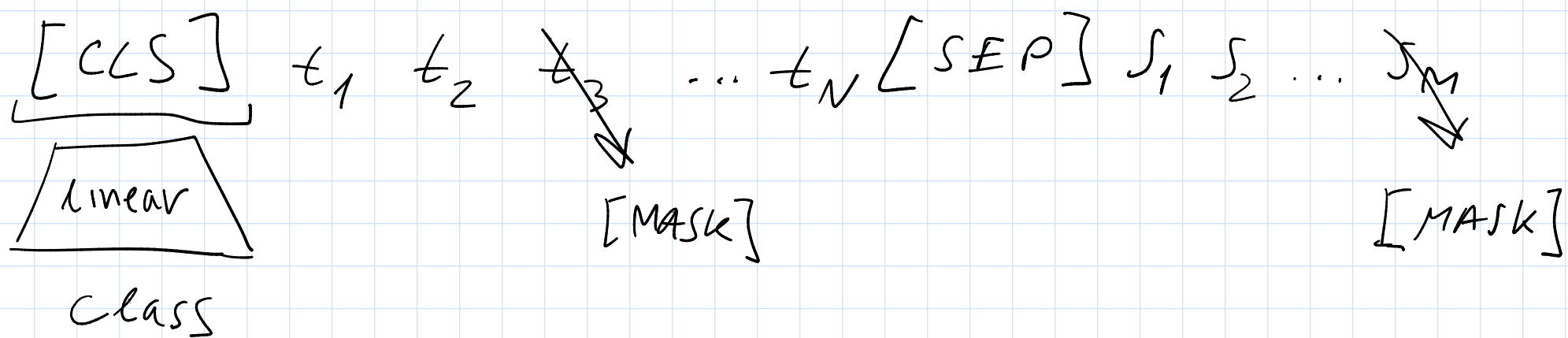
10%

unchanged

$$\text{BERT loss} = L_{\text{MLM}} + L_{\text{NSP}}$$

NSP - next sentence prediction

Model input



$$\begin{array}{cccccccccccccccc}
 E_{[CLS]} & E_{t_1} & E_{t_2} & \dots & E_{t_N} & E_{[SEP]} & E_{s_1} & \dots & E_{s_M} \\
 + & & & & & & & & \\
 E_0 & E_1 & E_2 & \dots & E_N & E_{N+1} & \dots & E_{N+M+1} \\
 + & & & & & & & & \\
 E_A & E_A & E_A & \dots & E_A & E_A & E_B & \dots & E_B
 \end{array}$$

GLUE (General Language Understanding Evaluation) (~2016)

• NLP

— CoLA - corpus of language acceptability

— MNLI - Multi-genre Natural Language Inference

• NLP

ROBERTA

ALBERT

• Images

BEiT

• Audio

HUBERT

SSL for images

Pre-text tasks:

— Rotation angle prediction

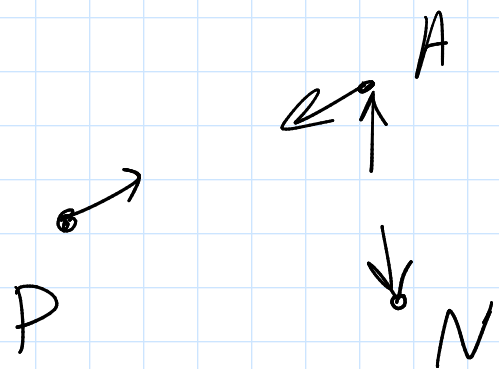
— Jigsaw puzzle

— Image colorization

grayscale $\rightarrow$ color

Triplet Loss

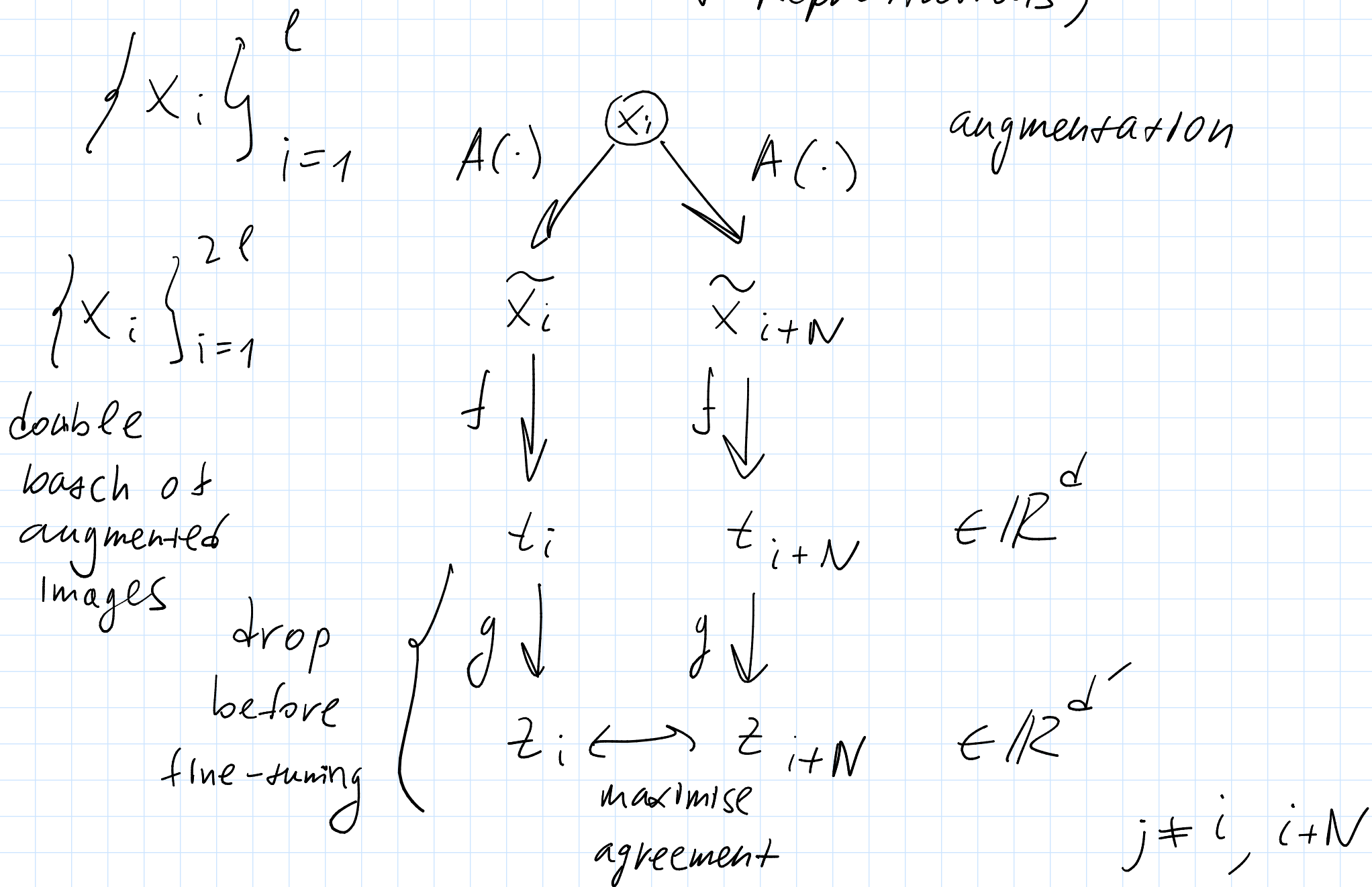
$$\|f(x) - f(y)\|_2^2 \quad \text{--- metric learning}$$



$$\max(\|f(A) - f(P)\| - \|f(A) - f(N)\| + m, 0) \rightarrow \min_f$$

Contrastive Learning

SimCLR (A simple Framework for Contrastive Learning of Representations)



Contrastive Loss

$$\ell(z_i, z_{i+N}) = -\log \frac{\exp(\text{sim}(z_i, z_{i+N}))}{\sum_{k=1}^{2N} \mathbb{1}\{k \neq i\} \exp(\text{sim}(z_i, z_k))}$$

$$\text{sim} = \frac{\langle x, y \rangle}{\|x\|_2 \|y\|_2}$$

$$L(z_i, z_{i+N}) = \frac{1}{2} (\ell(z_i, z_{i+N}) + \ell(z_{i+N}, z_i))$$

- Random Crop
- Color Jitter
- Grayscale
- Gaussian Blur